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Precision of the Schrödinger and Ondulatory (TauTimp) Wave Equations Against NIST Spectral Data: Hydrogen, Helium, Lithium, Beryllium, Boron, Carbon

Sources: `Heisenberg.docx` (uncertainty principle), `electron_TauTimp_final_RO.md` (TauTimp ondulatory equation), `Corectie_Relativista_TauTimp_v2_RO.md` (relativistic kinetic correction, this conversation), the Hydrogen–Lithium–Beryllium–Boron–Carbon equilibrium series (Heisenberg equilibrium, this conversation)
Experimental data: NIST Atomic Spectra Database, NIST Ground Levels and Ionization Energies
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1 Introduction and central conclusion

This document compares the precision of **two** theoretical formulations — the standard Schrödinger equation and the ondulatory (TauTimp) wave equation from `electron_TauTimp_final_RO.md` — against real experimental data (NIST): ionization energies and spectral lines, for the first six atoms of the periodic table: H, He, Li, Be, B, C. Compared to the initial version, it now includes an explicit quantification of the relativistic kinetic correction with effective mass $m_{\text{eff}} = m/\sqrt{1 - V^2/c^2}$ (the correct, standard form of relativistic mass — see `Corectie_Relativista_TauTimp_v2_RO.md`), applied to all six elements.

The TauTimp source document already establishes (sections 4.2–4.3, numerically verified there and reconfirmed here for every atom in the series) that **the modified TauTimp ODE, with the cosmological radius R held constant, is algebraically identical to the Schrödinger equation at**

any atomic scale, with a corrective cosmological tail of order $(r/R)^4 \lesssim 10^{-130}$ — completely unmeasurable. The consequence for this document:

> **The two theories are indistinguishable from one another when compared against NIST data.** Any error against experiment identified below is identical for Schrödinger and for TauTimp — neither is "more precise" at atomic scale. The real discrepancies from reality come from **the level of approximation of the atomic model used** (single hydrogenic electron vs. real multi-electron atom), not from the choice of wave equation.

2 Methodology

For **single-electron systems** (H, and the He^+ ion), the **exact** Schrödinger/TauTimp solution is used, with no screening approximation whatsoever — the comparison with NIST directly tests the accuracy of the wave equations themselves, including reduced-mass and relativistic-kinetic corrections.

For **neutral multi-electron atoms** (He, Li, Be, B, C), the screened hydrogenic model developed in the preceding documents of this series is used: Slater's rules for He/Li/Be, Clementi–Raimondi charges for B/C (more accurate, distinguishing $2s$ from $2p$). Here the comparison with NIST tests **both** the wave equation (identical for Schrödinger and TauTimp) **and** the quality of the independent-electron approximation with a fixed effective charge — the dominant source of error, as will become clear.

For the relativistic kinetic correction, we use the standard form of relativistic mass, $m_{\text{eff}}(V) = m/\sqrt{1 - V^2/c^2}$, inserted into the kinetic term of the energy functional (section "Quantifying the residual" details the derivation and closed form), and compare the result to the rigorous relativistic term $\hat{H}_{\text{rel}} = -\hat{p}^4/8m^3c^2$, already present in the extended TauTimp ondulatory equation (source document, §6.3).

3 Hydrogen — near-perfect agreement

3.1 Ground-state energy: infinite mass vs. reduced mass

$E_1(R_\infty) = -13.605693$ eV (the $Z_{eff} = 1$, electron-mass convention used throughout the series)

$$E_1(R_H) = -\mu c^2 \alpha^2/2 = -13.598287 \text{ eV} \qquad \mu = \frac{m_e m_p}{m_e + m_p} \text{ (electron–proton reduced mass)}$$

$$E_1^{\text{NIST}} = -13.598434 \text{ eV}$$

Model	Relative error vs. NIST
R_∞ (infinite mass, series convention)	$+0.0534\%$
R_H (reduced mass)	-0.0011%

The simple reduced-mass correction shrinks the error by **~50×**. The remaining $+0.147$ meV residual (R_H is slightly under-bound relative to reality) is quantitatively explained below.

3.2 Lyman and Balmer spectral lines

Computed with R_H (reduced mass), $hc = 1239.84$ eV·nm; compared to reference values (Lyman in vacuum; Balmer historically quoted in air, converted to vacuum with $n_{\text{air}} = 1.000277$):

Series	Transition	λ_{model} (vacuum, nm)	$\lambda_{\text{reference}}$ (vacuum, nm)	Relative error
Lyman α	$2 \to 1$	121.568	121.567	0.0012%
Lyman β	$3 \to 1$	102.573	102.572	0.0013%
Lyman γ	$4 \to 1$	97.255	97.254	0.0008%

Series	Transition	λ_{model} (vacuum, nm)	$\lambda_{\text{reference}}$ (vacuum, nm)	Relative error
Balmer α (Hα)	$3 \to 2$	656.470	656.461	0.0014%
Balmer β (Hβ)	$4 \to 2$	486.274	486.270	0.0009%
Balmer γ (Hγ)	$5 \to 2$	434.173	434.167	0.0013%
Balmer δ (Hδ)	$6 \to 2$	410.293	410.288	0.0013%

All seven lines agree with experiment to within $\sim 0.001\%$ — remarkably uniform precision across both series, once the air-to-vacuum conversion is applied to the Balmer lines.

3.3 Quantifying the residual: the relativistic kinetic correction $m_{\text{eff}} = m/\sqrt{1 - V^2/c^2}$

The standard relativistic mass, $m_{\text{eff}}(V) = m/\sqrt{1 - V^2/c^2} = \gamma m$, inserted into the kinetic term of the energy functional $\langle H \rangle_{1s}(\beta) = \hbar^2/2m\beta^2 - Z_{eff}ke^2/\beta$, gives (full derivation in `Corectie_Relativista_TauTimp_v2_RO.md`):

$$\boxed{\Delta E_{\text{rel}} \approx -|E_0| \cdot \frac{1}{2}(Z_{eff}\alpha/n)^2} \qquad \text{(heuristic mass substitution)}$$

Compared with the **rigorous** relativistic term, already present in the extended TauTimp ondulatory equation (source document, §6.3, $\hat{H}_{\text{rel}} = -\hat{p}^4/8m^3c^2$, derived from the exact energy-momentum relation):

$$\boxed{\Delta E_{\text{rigorous}} = -\frac{T_0^2}{2mc^2} = -|E_0| \cdot \frac{1}{4}(Z_{eff}\alpha/n)^2} \qquad \Rightarrow \qquad \frac{\Delta E_{\text{rel}}}{\Delta E_{\text{rigorous}}} = 2.000 \text{ (exact)}$$

Applying both to hydrogen, starting from R_H (already reduced-mass-corrected):

Correction applied on top of R_H	ΔE (meV)	Resulting prediction (eV)	Gap vs. NIST (meV)
(none)	—	-13.598287	$+0.147$
ΔE_{rel} (heuristic $m/\sqrt{1 - V^2/c^2}$ substitution)	-0.362	-13.598649	-0.215
$\Delta E_{\text{rigorous}}$ ($-\hat{p}^4/8m^3c^2$)	-0.181	-13.598468	$\boxed{-0.034}$

The rigorous formula shrinks the gap by $\sim 4.3\times$ (from 0.147 to 0.034 meV); the heuristic mass substitution, while having the correct physical sign (more binding, matching the Dirac formula), *overestimates* the correction by exactly a factor of 2 and actually **worsens** agreement with NIST for hydrogen. The final 0.034 meV residual is at the scale of the Lamb shift (4.375 μeV) and hyperfine structure, already formalized (fully or phenomenologically) in the source document (§8–9).

4 Helium — the contrast between the ion (one electron) and the neutral atom (two electrons)

4.1 He^+ — exact hydrogenic ion, no screening

$E(\text{He}^+, 1s, \text{reduced mass}) = -54.415354$ eV $\qquad I_2^{\text{NIST}}(\text{He}) = 54.41778$ eV $\qquad$ gap $= +2.427$ meV

Applying the same comparison as for hydrogen:

Correction applied	ΔE (meV)	Resulting gap (meV)
(none, reduced mass only)	—	$+2.427$
ΔE_{rel} (heuristic)	-5.795	-3.369 (worse)
$\Delta E_{\text{rigorous}}$	-2.897	$\boxed{-0.471}$ ($\sim 5\times$ better)

The same pattern as hydrogen: the rigorous formula significantly improves agreement; the heuristic mass substitution overestimates and worsens the prediction.

4.2 Neutral He — the screened Slater model

$$E_{\text{total}}(\text{He}, 1s^2, Z_{eff} = 1.70) = -78.6409 \text{ eV} \qquad -(I_1 + I_2)^{\text{NIST}} = -79.0052 \text{ eV}$$

$$\boxed{\text{Error} = -0.461\% \quad (3026 \text{ meV, absolute})}$$

The rigorous relativistic correction for this case ($\Delta E_{\text{rigorous}} = -1.513$ meV, next section) explains under 0.05% of this 3026 meV gap — going from one electron to two increases the error by **~2000×** relative to the available relativistic correction, the first direct manifestation of this document's central limitation: **the independent-electron approximation with a fixed effective charge cannot represent real electron–electron repulsion, and single-electron relativity cannot compensate for it.**

5 Lithium, Beryllium, Boron, Carbon — the systematic degradation of precision

5.1 Summary table, with the rigorous relativistic correction

Atom	Valence state	Z_{eff}	E_{model} (eV)	E^{NIST} (eV)	Relative error	$\Delta E_{\text{rigorous}}$ (meV)	Fraction of gap
H	$1s$	1	-13.5983	-13.5984	0.001%	-0.181	2.5%
He^+	$1s$	2	-54.4154	-54.4178	0.004%	-2.897	119.6%
He	$1s^2$	1.70	-78.6409	-79.0052	0.461%	-3.026	0.10%
Li	$2s^1$	1.30	-5.7484	-5.3917	6.62%	-0.032	0.009%
Be	$2s^2$	1.95	-12.9339	-9.3227	38.74%	-0.164	0.005%
B	$2p^1$	2.42	-19.9201	-8.2980	140.06%	-0.388	0.003%
C	$2p^2$	3.14	-33.5367	-11.2603	197.83%	-1.101	0.005%

(For H and He^+, "Fraction of gap" is computed against the residual remaining after the reduced-mass correction, not the raw Z_{eff}/R_∞ error; for the others, against the total $E_{\text{model}} - E^{\text{NIST}}$ gap.)

5.2 Physical explanation of the trend

The error grows **monotonically and nearly exponentially** with the number of electrons in the valence/near-valence ($n = 2$) shell. The cause: **the electron–electron repulsion energy (the Coulomb and exchange integrals of a full Hartree–Fock calculation) is not included** in the simple hydrogenic formula $E = -13.6057\, Z_{eff}^2/n^2$ — neither the reduced-mass nor the relativistic kinetic correction touches this cause, both being single-electron effects.

6 TauTimp vs. Schrödinger — the same error, by construction

For every case above, the cosmological tail $(r/R)^4$ of the modified TauTimp ondulatory equation remains, at all relevant equilibrium radii, between $\sim 10^{-132}$ and $\sim 10^{-134}$ — completely negligible,

over 130 orders of magnitude below any error reported above, including the relativistic kinetic correction (μeV–meV). **None of the errors in the preceding sections can be attributed to a difference between TauTimp and Schrödinger** — both theories, at atomic scale, produce identical predictions, whether relativistically corrected or not.

7 Ways to improve the atomic model

7.1 For single-electron systems (H, He^+, hydrogenic ions)

1. **Reduced mass** — shrinks hydrogen's error by $\sim 50\times$ (Hydrogen section).
2. **The rigorous relativistic kinetic correction**, $\widehat{H}_{\text{rel}} = -\hat{p}^4/8m^3c^2$ — shrinks the remaining residual by a further ~ 4–$5\times$ (Hydrogen, Helium sections). **Do not** use the heuristic mass substitution $m/\sqrt{1-V^2/c^2}$ as a quantitative replacement — it has the correct sign but overestimates by exactly $2\times$ and worsens the prediction if applied alone.
3. **Spin-orbit coupling and the Darwin term** — already formalized in the extended TauTimp ondulatory equation (source, §6–7); complete the kinetic correction into a full Dirac formula.
4. **Lamb shift and hyperfine structure** — at the μeV scale, already treated (phenomenologically and fully, respectively) in the source (§8–9).

Fully applied, these corrections bring the prediction for single-electron systems into agreement with experiment limited only by the precision of the fundamental physical constants.

7.2 For multi-electron atoms (neutral He, Li, Be, B, C)

The corrections above **do not solve** the 0.46%–198% errors — as the Lithium–Carbon table clearly shows, the rigorous relativistic correction explains under 0.1% of any of these gaps. Effective directions:

1. **Self-consistent field (Hartree–Fock)** — replaces the fixed Z_{eff} with a self-consistently computed mean potential.
2. **Explicit electron correlation** (CI, Møller–Plesset, coupled-cluster CCSD(T), or Hylleraas for helium) — closes essentially all the gap remaining after Hartree–Fock.
3. **Antisymmetrized multi-electron wavefunction** (Slater determinant).
4. **Systematic, self-consistent relativistic corrections** for heavier atoms (Dirac–Fock) — useful only after correlation is solved, not before.

7.3 A specific extension of the TauTimp ondulatory equation

The TauTimp ondulatory equation, with or without the relativistic kinetic correction tested here, remains a **single-electron** equation. To surpass (not merely match) Schrödinger's precision for multi-electron atoms, TauTimp would need an explicit extension, currently absent from the source: coupling multiple ondulatory fields with electron–electron interaction terms, or a self-consistent Hartree–Fock-like loop built on the TauTimp formalism — remaining, like the first-principles derivation of the Lamb shift, an **open research program**.

8 Conclusions

1. **Schrödinger and TauTimp are experimentally indistinguishable at atomic scale** — even after adding the relativistic kinetic correction, the cosmological tail remains $\lesssim 10^{-130}$ relative.

2. **For single-electron systems** (H, He^+), the rigorous relativistic kinetic correction ($-\hat{p}^4/8m^3c^2$, already in the source document) shrinks the residual remaining after reduced mass by a factor of 4–5; the heuristic mass substitution $m/\sqrt{1-V^2/c^2}$ has the correct physical sign but overestimates by exactly $2\times$ and should not be used quantitatively.

3. **For multi-electron atoms**, the relativistic correction — rigorous or heuristic — explains under 0.1% of the 0.46%–198% gaps against NIST, quantitatively confirming that single-electron relativity is not the missing mechanism.

4. **The most effective improvement path** for atoms B–C remains a self-consistent multi-electron treatment with explicit electron correlation — a methodological extension, not a change to the underlying wave theory.

9 Appendix: Experimental data (NIST) and constants used

Quantity	Value	Source
I_1(H)	13.598434 eV	NIST
I_1(He)	24.58741 eV	NIST
I_2(He)	54.41778 eV	NIST
I_1(Li)	5.3917 eV	NIST
I_1(Be)	9.3227 eV	NIST
I_1(B)	8.29803 eV	NIST
I_1(C)	11.26030 eV	NIST
Lyman α, β, γ (vacuum)	$121.567; 102.572; 97.254$ nm	standard spectroscopic data
Balmer $\alpha, \beta, \gamma, \delta$ (air)	$656.279; 486.135; 434.047; 410.174$ nm	standard spectroscopic data
n_{air} (visible)	1.000277	standard refractive index
R_∞	13.605693 eV	CODATA
R_H (reduced mass)	13.598287 eV	computed, $\mu = m_e m_p/(m_e + m_p)$
α (fine structure)	$7.2973525693 \times 10^{-3}$	CODATA
ΔE_{rel} (heuristic)	$-\lvert E_0\rvert (Z_{eff}\alpha/n)^2/2$	$m_{\text{eff}} = m/\sqrt{1 - V^2/c^2}$
$\Delta E_{\text{rigorous}}$	$-\lvert E_0\rvert (Z_{eff}\alpha/n)^2/4 = -T_0^2/2mc^2$	$\widehat{H}_{\text{rel}} = -\hat{p}^4/8m^3c^2$
R (TauTimp cosmological radius)	4.2355×10^{22} m	`electron_TauTimp_final_RO.md`

This document evaluates the shared precision of the Schrödinger equation and the TauTimp ondulatory equation against real experimental (NIST) data, for H, He, Li, Be, B, and C, now including an explicit quantification of the relativistic kinetic correction with effective mass $m/\sqrt{1 - V^2/c^2}$, compared against the rigorous relativistic term already present in the source document. The central conclusion stands: the two theories are indistinguishable at atomic scale; the real error for multi-electron atoms comes from the level of approximation of the model (fixed effective charge, no electron correlation), not from the choice of wave equation nor from single-electron relativistic corrections.

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